Assignment 1: Basics of Graph Plotting

Summer Break Bridge Course

Exercises

1. Even, odd, or neither

Check whether each of the following functions is even, odd, or neither. If it is neither, express it as a sum of an even part and an odd part, clearly showing which part is even and which part is odd.

(a) $\sin(|5x|)$

(b) $\tan(x/6)$

(c) $5x^2 + 3x$

(d) e^{-3x}

(e) $f(x) = -x + x^2 - x^3 + x^4 - x^5 + \cdots$

2. Hand sketch first, then Python plot

For each of the following, first roughly sketch all the functions in that part on the same graph by hand. Clearly indicate:

- the value at $x = 0$, whenever defined,
- the value at $x = 1$, whenever defined,
- and the behavior as $x \to \pm\infty$, whenever applicable.

Then plot the same functions in Python. In each part, use the same `x_array` for all functions in that part. Label each function clearly and include a legend.

(a) e^{x}, e^{3x}, $e^{x/2}$
For numerical plotting, take 500 x-values between -4 and 4.

(b) e^{-x}, e^{-3x}, $e^{-x/2}$
For numerical plotting, take 500 x-values between -4 and 4.

(c) $\frac{1}{x}$, $\frac{1}{x-2}$, $\frac{1}{x+5}$
For numerical plotting, create separate x-arrays with `x_array_1` having values $\in [-8, -5)$ excluding $x = -5$, with spacing of 0.01, `x_array_2` having values $\in [-5.01, 0)$ excluding $x = 0$, with spacing of 0.01, `x_array_3` having values $\in [0.01, 2)$ excluding $x = 2$, with spacing of 0.01, `x_array_4` having values $\in [2.01, 8]$, with spacing of 0.01. Then combine them using `np.concatenate` by writing like this -
`x_array = np.concatenate((x_array_1, x_array_2, x_array_3, x_array_4))`.
Actually, `np.concatenate` joins multiple arrays together to make a new array. Use the final `x_array` for calculating y-values and plotting.

(d) $\sin x,\ \sin(4x),\ \sin(x+1),\ \sin x + 0.5$
For numerical plotting, take 500 x-values between -4 and 4.

(e) $\ln x,\ \ln(3x),\ \ln(x+2)$
For numerical plotting, take x-values from 0.01 to 4 with spacing 0.005.

(f) $x^3,\ (x-2)^3,\ x^3-2$
For numerical plotting, take 1000 x-values between -3 and 3.